

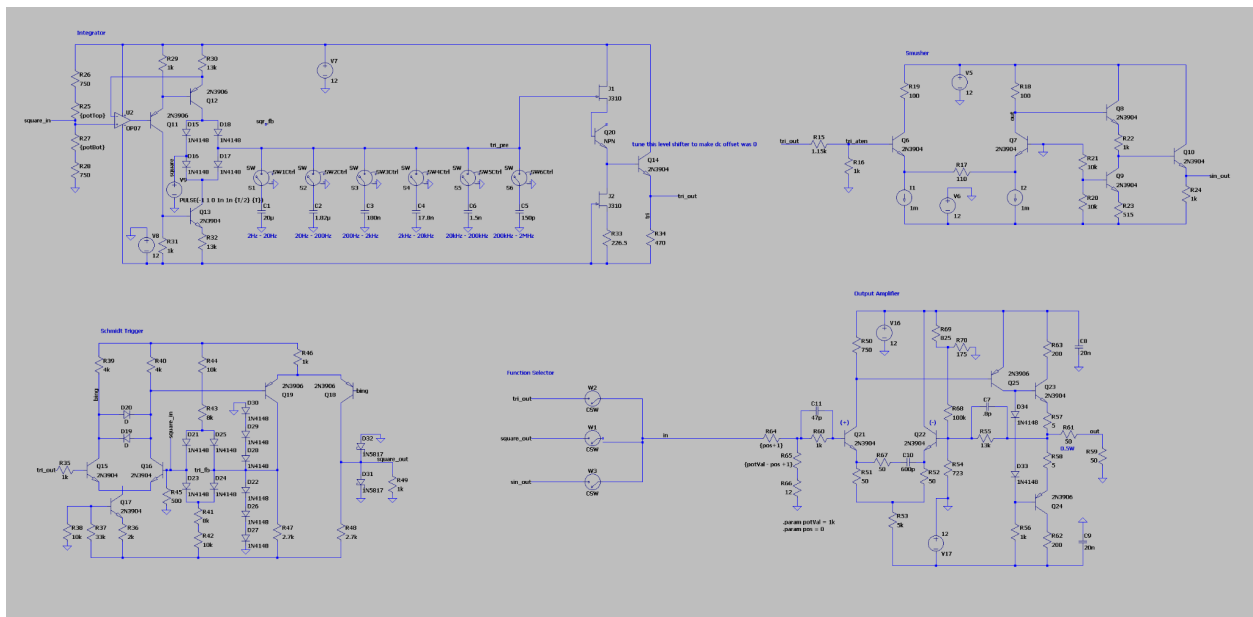
Lab 4 Writeup

Adri and Finn

Achieved Specs

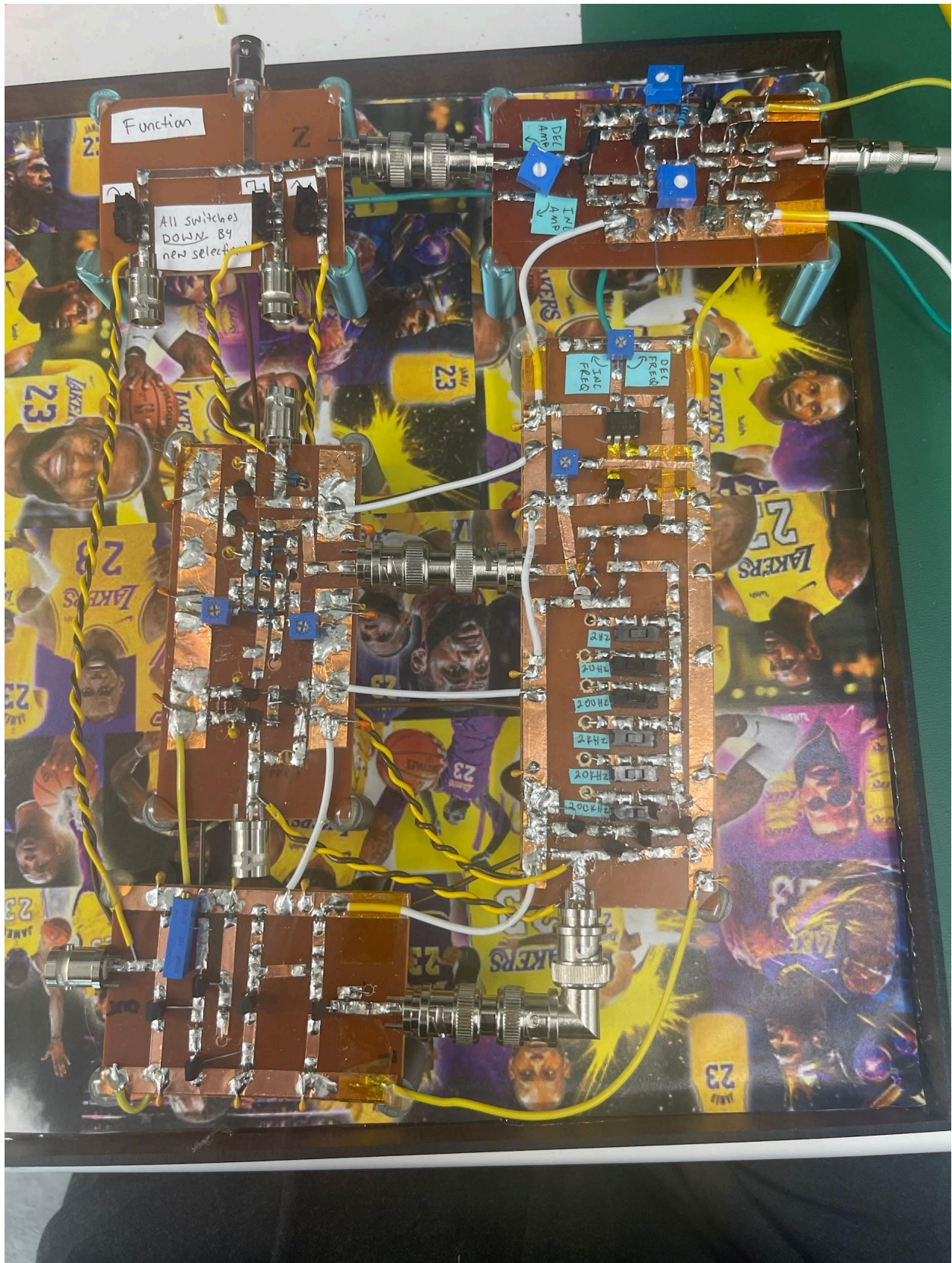
Spec	Goal	Achieved
THD	<3%	2.6%
Max Amplitude	>3.2V	8V
Frequency Range	2Hz - 2MHz	1.9Hz - 2.2MHz
Square Overshoot	<10%	2%
Duty Cycle Error	<2%	4%
Worst Case Triangle Peak Variation	<10%	4%
Output Short to Ground	Survive 10s Short to Ground at Max Amplitude	Yes! It survives the 10s short at 8V Amplitude

Final Schematic

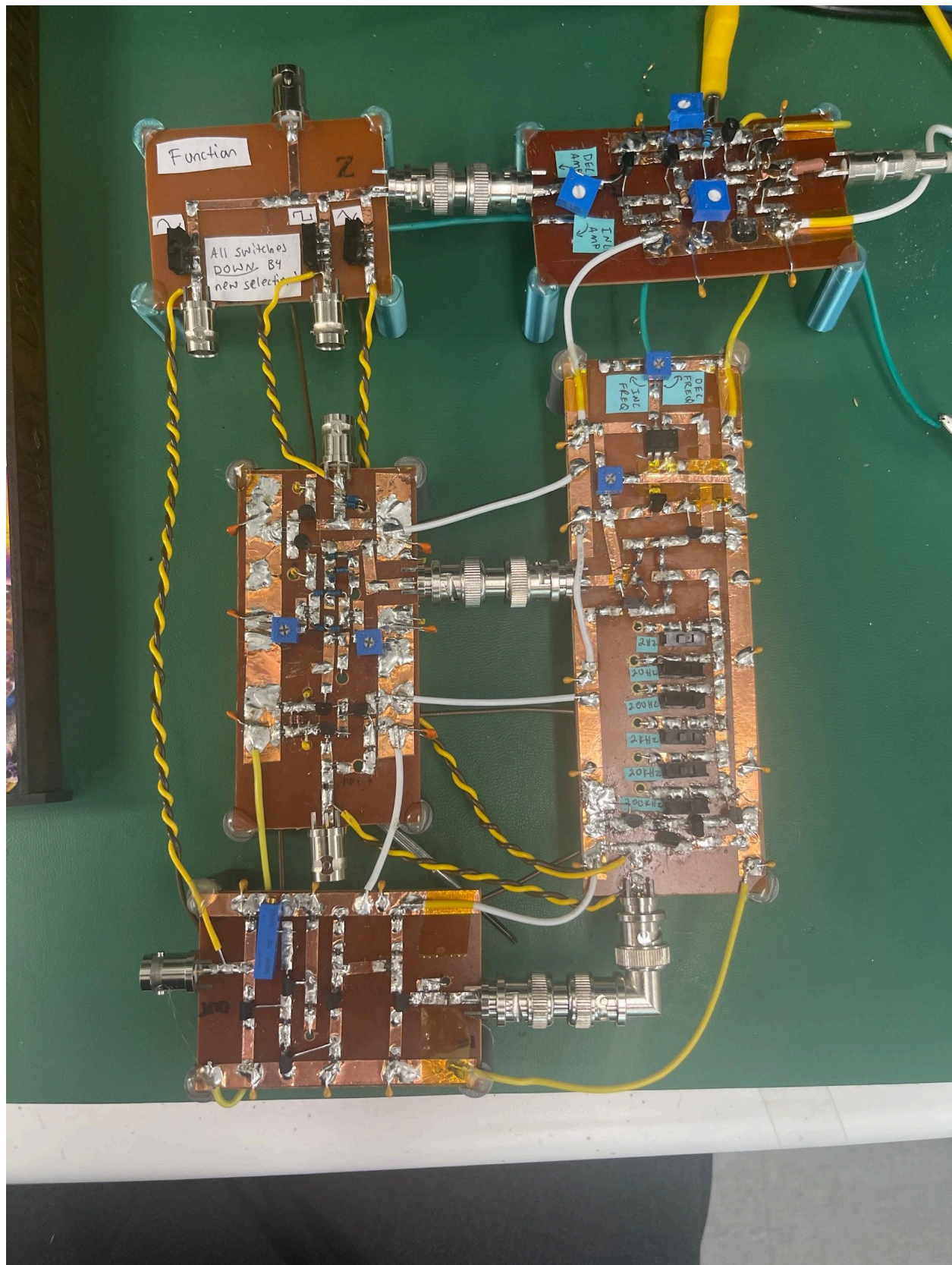


Physical Design Picture

With LeBron James



Without LeBron James



Discussion

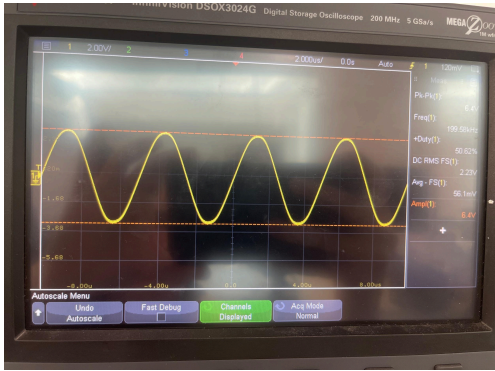
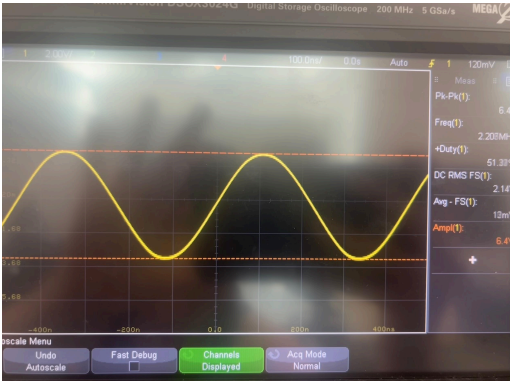
Our final board was able to achieve most of the specs, including having continuous function across a frequency range from 2Hz - 2Mhz. Although it was slightly mistuned in the demo with proper tuning we have no gaps as can be seen below. We can modulate the amplitude of our wave from $<0.5V$ to $3.2V$, as well as above that up to $8V$. For the square wave there is minimal overshoot ($<10\%$), and we have a rise time of about $26nS$ and a fall time of $29nS$, meaning we have decent bandwidth and slew rate. The DC offset for both the square and triangle wave can be tuned to be $<100\text{ mV}$, although the triangle wave still has a bit of an offset which is discussed next. We are able to short our design at the maximum output voltage amplitude $8V$, and have the signal return after removing the short.

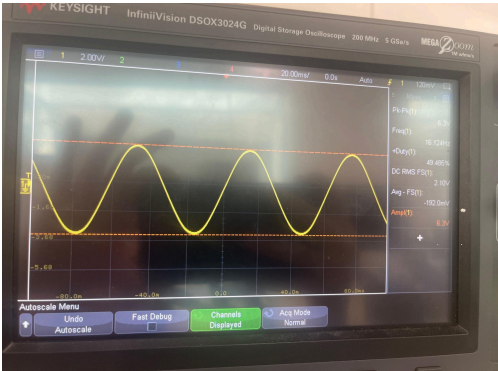
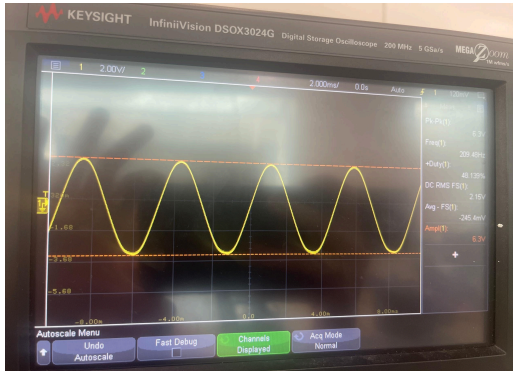
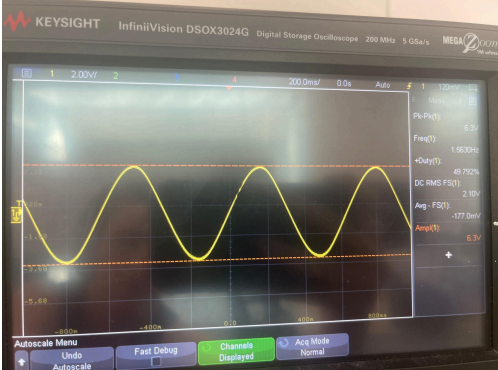
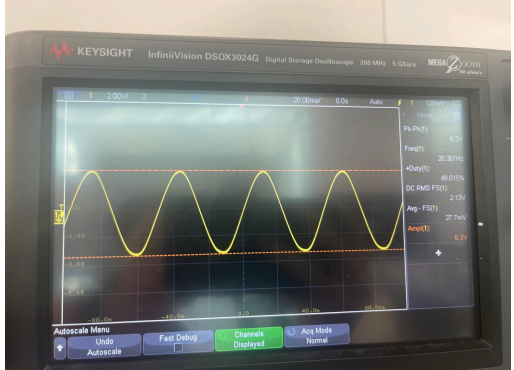
There were a couple of specs we did not quite meet, which were discussed in the demo and can be seen in the waveforms. We had issues with the symmetry of our wave, which caused our duty cycle to vary above 2% across the spectrum. This was due to us attempting to pull more current than the op-amp could provide for our symmetry network, prevent it from being very adjustable. Additionally, we seem to have a small artifact or noise interfering with our triangle and square wave at lower amplitudes, which can be seen pictured below. Finally, we have a stubborn DC offset on our triangle wave that is slightly out of spec ($200mV$). This is from the JFet used in the output buffer. A way to fix this could be to potentially replaced the JFET current source we currently have at the output of the integrator with a BJT current source that we can tune the current more precisely to have a specific voltage drop across a collector resistor to tune out the offset (similar to what was implemented in the sin smusher to tune out the sine offset).



Scope Captures

Sine

Range	Lower	Upper
2-20Hz		
20-200Hz		
0.2-2kHz		

2-20kHz		
20-200kHz		
0.2-2MHz		

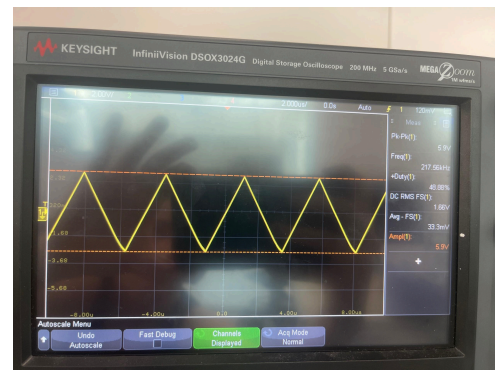
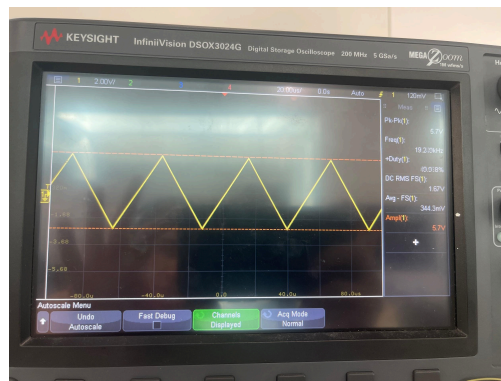
Triangle

Range	Lower	Upper
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2-20Hz



20-200Hz

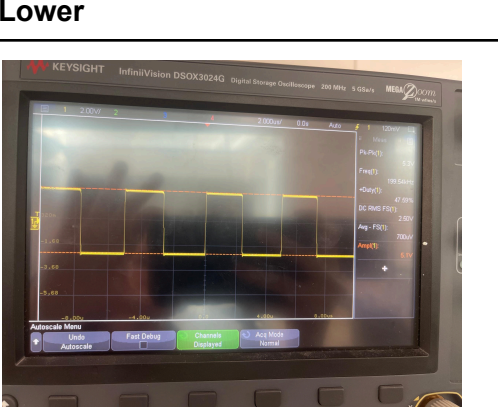
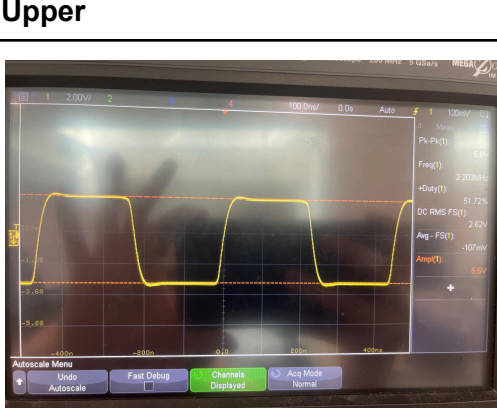


0.2-2kHz



2-20kHz



20-200kHz		
0.2-2MHz		
Square	Range	Upper
2-20Hz		

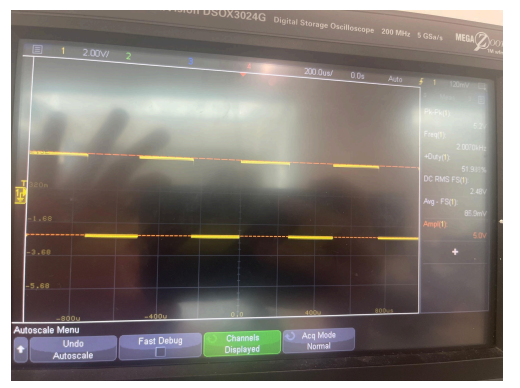
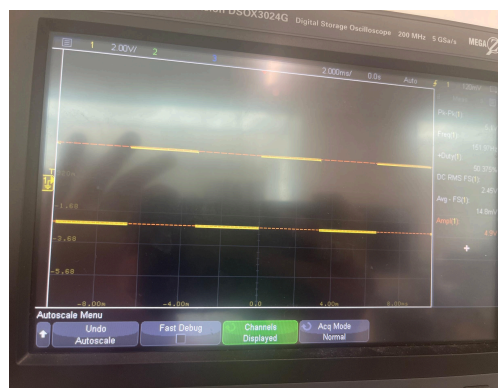
20-200Hz



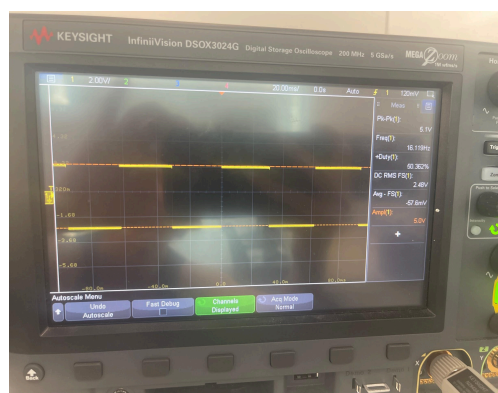
0.2-2kHz



2-20kHz



20-200kHz



0.2-2MHz

